



Nonequilibrium dynamics in a noise-induced predator–prey model

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ABSTRACT

Understanding the dynamics of predator–prey systems in the presence of different environmental variability is crucial in ecology for forecasting population behaviour and ensuring ecosystem sustainability. It is a challenging aspect of studying spatio-temporal dynamics in the presence of environmental variability. We provide a noise-induced spatio-temporal predator–prey model to explore how temporal variability and spatial heterogeneity affect population dynamics. The multiplicative stochastic fluctuations in space and time are considered in the prey's growth rate and predator's death rate to capture the demographic noise in the ecosystem. We first examine the deterministic models by finding crucial parameters that influence the stability and dynamics of predator and prey populations, following their impact on spatio-temporal pattern formation. Using analytical tools and numerical simulations, we illuminate the mechanisms behind the observed dynamics and highlight the significance of demographic noise in generating ecological patterns. The numerical simulations show that the temporal variability introduced by noise leads to oscillations in population densities and alters the stability of the predator–prey system. Special attention is given to the spatio-temporal system when it fails to produce Turing patterns without noise, and the results show that linear demographic change can cause complex behaviours, such as self-organization, irregular oscillations, and nonequilibrium dynamics. Nevertheless, these findings broadly affect various ecological phenomena, including population persistence, species coexistence, long transients, and ecosystem resilience to demographic perturbations.

1. Introduction

Population dynamics is a multifaceted discipline of ecology that investigates complex patterns of change in organismal populations over time and across locations. It investigates the relationships between birth rates, death rates, immigration, and emigration, all of which influence the density, distribution, and composition of populations within ecosystems [1]. These dynamics are impacted by various variables, including resource availability, predation, competition, climatic fluctuation, and human activity [2,3]. Understanding population dynamics is critical for understanding ecosystem resilience, forecasting species responses to environmental change, and directing conservation and management initiatives [4–6]. Ecologists use a variety of methods, including mathematical models, empirical research, and long-term monitoring, to try to gain insight into the fundamental processes that cause population fluctuations and come up with plans for the sustainable management of natural resources and biodiversity in an environment [7,8].

A predator–prey model is a mathematical representation used in ecology to study their interactions in an ecosystem [9,10]. This modelling technique attempts to explain population dynamics across time

by anticipating how changes in one species' abundance impact the other, and vice versa [11]. The standard predator–prey model consists of equations that describe the population densities of a predator species and its prey species. In the early twentieth century, Alfred Lotka and Vito Volterra separately developed one of the most well-known predator–prey models called the Lotka–Volterra equations [12,13], which serve as the foundation for several future studies on population dynamics. These models are frequently called a system of coupled differential equations considering birth and death rates, predation, and resource competition. For instance, this paper considers an extension of the Lotka–Volterra equation [14], which captures some complexities such as functional response, density-dependent death rate, etc. While these deterministic models give valuable insights, they have drawbacks, such as assuming constant parameters and failing to account for environmental unpredictability or geographical concerns [15]. More comprehensive models have been created to overcome these shortcomings that incorporate stochasticity and other ecological subtleties, allowing them to better depict the complexities of predator–prey interactions in varied ecosystems.

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Noise-induced predator–prey dynamics is an ecological concept that emphasizes how demographic fluctuations can substantially impact predator–prey interactions within an ecosystem [16]. It can occur from various causes, including environmental conditions, demographic stochasticity, and other unexpected impacts [17,18]. The authors in [17] showed that multiplicative noise is responsible for two species' coherent oscillations, whereas additive noise regulates the transition between the coexistence and exclusion dynamical regimes. Meanwhile, the traditional deterministic models presume stable parameters and predictable interactions, where populations follow smooth trajectories or exhibit regular oscillations. However, environmental unpredictability and population randomization in real-world ecosystems add noise to these processes, which are the unavoidable component that impacts species models [19]. These noises can come from different sources, such as temperature, light, and nutrition availability. This unpredictability can destabilize populations because certain species may encounter a periodic lack of resources, resulting in swings that raise extinction risks, especially in smaller populations.

Noises can be characterized as additive or multiplicative, depending on their source. For the additive noise, the random fluctuations are added directly to both prey and predators at each time step to take care of the random events like weather fluctuations affecting the populations. On the other hand, for the multiplicative noise, the demographic fluctuations affect the interaction rates within the system, which can come from resource availability or predation efficiency. These noises might have unanticipated consequences, such as irregular oscillations, transient behaviour, or a generation of new stable states [20]. These noise terms make the model more realistic by accounting for natural ecosystems' inherent uncertainty and variability. In addition, it can enhance the chance of population extinctions or contribute to the existence of alternative stable states, where the system can exhibit different stable patterns of population dynamics under the same environmental conditions [21,22].

This work focuses on how linear demographic change influences the temporal and spatio-temporal behaviour of predator–prey systems. Here, we have considered multiplicative noise in the stochastic model to incorporate the linear dependence of the growth and death rates for the prey and predator populations, respectively, and studied its impact on the resulting dynamics. The Kolmogorov–Fokker–Planck (KFP) equation in stochastic processes is fundamental for demonstrating the development of probability distributions in systems affected by random fluctuations [23]. It captures the equilibrium between deterministic drift and stochastic diffusion, illustrating the system's interaction between deterministic dynamics and random fluctuations. Furthermore, the equation describes how randomness affects the development of physical systems, providing insights into collective behaviours, equilibrium states, and dynamical process stability [24]. In particular, the authors in [24] showed that the probability of noise-induced transitions between various attractors is crucial for cell differentiation and developmental processes. Bashkirtseva and Ryashko in [25] have shown that the system transitions from coexistence to extinction after a critical threshold of the noise. The stochastic sensitivity analysis has been applied to find such a critical threshold. Here, we have also applied the same approach to approximate the quasipotential for the steady-state probability distributions of the KFP equation around the stable attractors [25–27]. This gives us confidential ellipses around an exponentially stable attractor for different noise intensities. In addition, we have estimated the confidential bands for the stable limit cycle for different noise intensities. We have shown that the stochastic trajectories are well inside these confidential ellipses and bands for the lower noise intensity but can go outside for a higher noise intensity.

This study provides several intriguing facts about spatio-temporal patterns in the absence and presence of noise terms. We have demonstrated that the small perturbation of the homogeneous steady state through oscillatory modes can be suppressed completely to generate stationary patches in the presence of diffusion without noise. On the

other hand, the noise-induced model extends the parametric region of getting such patches. In addition, demographic perturbations and nonlinearity can help form patches in the system when their absence fails to do so, and this is one of the main goals of this work. Furthermore, the noise-induced pattern transitions between coexisting patterns are observed in the noise-induced model, showing the species' self-organization and being more resilient to the fluctuations. We have calculated the modality coefficients to analyse the qualitative shifts in long transients for the spatio-temporal model [28–30], and they show significant qualitative changes in patterns in the presence of noise.

2. Mathematical models

The basic Lotka–Volterra predator–prey model is widely used in ecology to understand the population dynamics of interacting species [12,13]. Verhulst incorporated logistic growth into the model to represent population expansion over time while accounting for limiting variables that restrain population size [31]. Rosenzweig and MacArthur extended the model to capture the complexities of ecological interactions, such as prey switching and functional responses, which provides a more realistic representation of predator–prey dynamics [32]. The latter concept is used for the first time by Michaelis and Menten for enzyme kinetics [33]. The functional response describes how an individual predator's consumption rate changes with prey density, while the numerical response refers to how the predator population density changes in response to prey availability due to the birth of new offspring. In this work, we consider an extension of the Rosenzweig–MacArthur model by incorporating a density-dependent death rate in predators by the following [14], known as Bazykin's model:

$$\begin{aligned}\frac{du}{dt} &= ru \left(1 - \frac{u}{K}\right) - \frac{auv}{b+u}, \\ \frac{dv}{dt} &= \frac{eauv}{b+u} - mv - qv^2,\end{aligned}\quad (1)$$

with non-negative initial conditions. Here, $u(t)$ and $v(t)$ are the densities of the prey and predator populations at time $t \in (0, T]$ for some $T > 0$, respectively. The parameters r and K represent the biotic potential and carrying capacity for the prey population, respectively, a is the consumption rate of prey by predators, and b is the half-saturation constant used for the first time in enzyme kinetics [33]. This half-saturation constant represents the substrate concentration at which the uptake or growth rate is half its maximum value, and it allows for more balanced nutrient utilization and reduces the risk of dominance by a single species. The parameter $e \in (0, 1)$ represents the conversion efficiency of the prey biomass into predator biomass, and m and q represent the intrinsic mortality rate and the intra-specific competition for the predators. This Bazykin model is an extension of the classical Rosenzweig–MacArthur model that introduces a density-dependent death rate among predators. All the parameters used here are assumed to be positive and do not vary with time.

Different environmental variability can be introduced into the model (1) to capture the effects of external factors, environmental variations, or demographic stochastic events that may impact the populations of prey and predators. These disturbances can be modelled in various ways, and one common approach is to include multiplicative noise terms in the equations to incorporate linear perturbation of the intrinsic growth and death rates for the prey and predator populations, respectively. We introduce such random noises in the system (1) by perturbing the parameters r and m as the noise terms $r + \sigma_1 \xi(t)$ and $m + \sigma_2 \eta(t)$ and obtain:

$$\begin{aligned}\frac{du(t)}{dt} &= F_1(u(t), v(t)) + \sigma_1 u(t) \xi(t), \\ \frac{dv(t)}{dt} &= F_2(u(t), v(t)) + \sigma_2 v(t) \eta(t),\end{aligned}\quad (2)$$

where $\xi(t)$ and $\eta(t)$ are the uncorrelated standard Gaussian noises with $E[\xi(t)] = E[\eta(t)] = 0$, $E[\xi(t)\xi(\tau)] = \delta(t - \tau)$, and $E[\eta(t)\eta(\tau)] = \delta(t - \tau)$ with δ is the Dirac delta function. Here, σ_1 and σ_2 are the environmental

forcing intensities, and we choose $\sigma_1 = \sigma_2 = \sigma$ for the simplicity of rigorous mathematical calculations. In addition, the reaction terms are $F_1(u, v) = ru(1 - u/K) - auv/(b + u)$ and $F_2(u, v) = eauv/(b + u) - mv - qv^2$. Furthermore, we extend the stochastic model (2) to the spatio-temporal noise-induced model as [34]:

$$\begin{aligned}\frac{\partial u(x, t)}{\partial t} &= d_u \frac{\partial^2 u(x, t)}{\partial x^2} + F_1(u(x, t), v(x, t)) + \sigma u(x, t) \xi(x, t), \\ \frac{\partial v(x, t)}{\partial t} &= d_v \frac{\partial^2 v(x, t)}{\partial x^2} + F_2(u(x, t), v(x, t)) + \sigma v(x, t) \eta(x, t),\end{aligned}\quad (3)$$

having non-negative initial and no-flux boundary conditions over a spatial domain $[0, L] \subset \mathbb{R}$. The parameters d_u and d_v are the diffusion coefficients for the prey and predator populations, respectively. In addition, the stochastic disturbances satisfy the properties: $E[\xi(x, t)] = E[\eta(x, t)] = 0$, $E[\xi(x, t)\xi(y, \tau)] = \delta(x - y)\delta(t - \tau)$, and $E[\eta(x, t)\eta(y, \tau)] = \delta(x - y)\delta(t - \tau)$ [29,35].

3. Near-equilibrium analysis

Linear stability analysis is a tool for understanding the behaviour of a deterministic dynamical system around its equilibrium points, and the system's dynamics depend on the combination of the behaviours of all the equilibrium points over the feasible domain of definition. The basic idea of this methodology is to linearize the system around an equilibrium point and examine the eigenvalues of the resulting linearized matrix. Understanding the stability properties of predator-prey interactions is crucial for managing ecosystems, predicting population dynamics, and informing conservation efforts. Furthermore, linear stability analysis identifies critical parameters that influence the persistence and resilience of predator and prey populations, which aids in developing effective management strategies to reduce the risks of over-exploitation, extinction cascades, and ecological collapse.

3.1. Linear stability analysis for the temporal deterministic system

The equilibrium points for the system (1) satisfy the algebraic equations $F_1(u, v) = 0$ and $F_2(u, v) = 0$. This shows that $E_0 = (0, 0)$ and $E_1 = (K, 0)$ are the trivial and axial equilibrium points, respectively. In addition, for a non-trivial coexisting equilibrium point $E_* = (u_*, v_*)$ of the system (1), u_* satisfies the following cubic polynomial equation:

$$P_3(u) \equiv qru^3 + (2bqr - qrK)u^2 + (a^2eK - 2bqrK + b^2qr - amK)u - ambK - b^2qrK = 0, \quad (4)$$

and v_* can be determined in terms of u_* using the relation $v_* = (eau_*/(b + u_*) - m)/q$. For the feasibility of the coexisting equilibrium point, u_* and v_* have to be positive. The cubic polynomial Eq. (4) has at least one positive root $u = u_*$ for all positive parameter values because the leading coefficient and the constant term are of opposite signs. Furthermore, Eq. (4) can have three distinct positive real roots if the derivative of $P_3(u)$ has two positive real roots, say u_1 and u_2 , and the corresponding values $P_3(u_1)$ and $P_3(u_2)$ have opposite signs. On the other hand, v_* is positive only when $u_* > bm/(ea - m)$ with $a > m/e$. Therefore, the existence of the non-trivial equilibrium point depends on the parameter values. Following the bifurcation theory, a transcritical bifurcation at E_1 generates the coexisting equilibrium point. In addition, saddle-node bifurcations are possible, which can occur for different combinations of the parameter values. We have discussed these transcritical and saddle-node bifurcations in Section 4 for a fixed set of parameter values except the parameter a . Here, we consider a as the bifurcation parameter that incorporated the grazing rate of predators, which is an important parameter involved with the functional response, the sole link between two trophic levels. Nevertheless, it is interesting to see the bifurcation of interior equilibrium from axial equilibrium through transcritical bifurcation and the spontaneous generation of coexistence equilibrium through saddle-node bifurcation.

For each equilibrium point, we find the Jacobian matrix as

$$\mathbf{J} = \begin{bmatrix} \frac{\partial F_1}{\partial u} & \frac{\partial F_1}{\partial v} \\ \frac{\partial F_2}{\partial u} & \frac{\partial F_2}{\partial v} \end{bmatrix} \equiv \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \quad (5)$$

and compute its eigenvalues. These eigenvalues help to determine the stability behaviour of the equilibrium point, whether the system converges to it or goes away from it. In the considered model (1), the trivial equilibrium point E_0 is always a saddle as the eigenvalues of the Jacobian matrix are r and $-m$. In addition, the axial equilibrium point E_1 is stable only when the coexisting equilibrium point does not exist. Furthermore, the stability of a coexisting equilibrium point depends on the eigenvalues of the corresponding Jacobian matrix, and they are implicitly involved in the roots of Eq. (4). In the next section, we have demonstrated these stabilities for a fixed set of parameter values by considering a as the bifurcation parameter.

3.2. Stochastic sensitivity analysis for the noise-induced temporal system

The linear stability shows whether a system returns to equilibrium after a small disturbance. However, it does not give us any information in the presence of random variability whether the states of the system stay around the equilibrium. So, there is a chance that the trajectories for the stochastic system (2) may leave the basin of attraction of the deterministic system in the presence of environmental forcing, and it depends on the magnitude of noise intensity. These trajectories form a stochastic attractor with a stationary probability distribution $P_{ss}(u, v, \sigma)$, and the Kolmogorov-Fokker-Planck (KFP) equation helps to find such distribution [23]. For the stochastic differential Eq. (2), the KFP equation is given by

$$\frac{\partial p}{\partial t} = -\frac{\partial(F_1 p)}{\partial u} - \frac{\partial(F_2 p)}{\partial v} + \frac{1}{2} \frac{\partial^2(G_1 p)}{\partial u^2} + \frac{1}{2} \frac{\partial^2(G_2 p)}{\partial v^2}, \quad (6)$$

where $G_1 = (\sigma u)^2$ and $G_2 = (\sigma v)^2$. Here, $p(\mathbf{z}, t)$ is the probability density function at $\mathbf{z} = (u, v)$ and time t [23,24]. The initial condition is $p(\mathbf{z}, 0) = \delta(\mathbf{z} - \mathbf{z}_*)$, where δ is the Dirac-delta function and $\mathbf{z}_* = (u_*, v_*)$. The no-flux boundary condition is considered over the boundary. Solving Eq. (6) analytically is not an easy task, so we use the asymptotic based on the quasipotential $\phi(\mathbf{z}) = -\lim_{\sigma \rightarrow 0} \sigma^2 \log P_{ss}(\mathbf{z}, \sigma)$ [25,36], and we can write $P_{ss}(\mathbf{z}, \sigma) \approx K \cdot \exp(-\phi(\mathbf{z})/\sigma^2)$ for a low noise intensity.

We use the quadratic approximation for the quasipotential $\phi(\mathbf{z}) = \frac{1}{2}(\mathbf{z} - \mathbf{z}_*, \mathbf{W}^{-1}(\mathbf{z} - \mathbf{z}_*))$ when the stochastic states of the noise system concentrate around an exponentially stable equilibrium point E_* , where the notation $(\cdot, \cdot)$ is a scalar product. The matrix $\mathbf{W}$ is called a stochastic sensitivity function, which can be obtained by solving the matrix equation:

$$\mathbf{J}\mathbf{W} + \mathbf{W}\mathbf{J}^T + \mathbf{S} = \mathbf{0}, \quad (7)$$

where $\mathbf{J}$ is the Jacobian matrix for the deterministic system at the equilibrium E_* and $\mathbf{S} = \text{diag}\{u_*^2, v_*^2\}$. The matrix $\mathbf{W}$ helps to approximate the stationary distribution of random states for the noise-induced system in the Gaussian form

$$P_{ss}(\mathbf{z}, \sigma) \approx K \cdot \exp(-(\mathbf{z} - \mathbf{z}_*, \mathbf{W}^{-1}(\mathbf{z} - \mathbf{z}_*)) / (2\sigma^2)).$$

Furthermore, the confidence ellipse for the noise intensity σ can be written as:

$$(\mathbf{z} - \mathbf{z}_*, \mathbf{W}^{-1}(\mathbf{z} - \mathbf{z}_*)) = 2s^2\sigma^2, \quad (8)$$

where $s^2 = -\ln(1 - \mathcal{P})$ with $\mathcal{P}$ as the fiducial probability (the probability of the random states for the stochastic system belong to this confidence ellipse).

For some parameter values, the deterministic system has no exponentially stable point; however, an exponentially stable limit cycle exists. In this case, we can construct a confidence band around the deterministic stable limit cycle similar to the stable attractor. We first suppose that $\mathbf{z} = \xi(t)$ is the T -periodic stable limit cycle for the deterministic system and Θ_t is the hyperplane orthogonal to the cycle

$\xi(t)$ where $0 \leq t < T$. In the presence of low-noise intensities, the stochastic attractor lies in the Poincare section Θ_l in the neighbourhood of $\mathbf{z} = \xi(t)$. In this case, the quadratic approximation of the quasipotential can be written as $\phi_l(\mathbf{z}) \approx \frac{1}{2}([\mathbf{z} - \xi(t)]^\top \mathbf{W}^+(t)[\mathbf{z} - \xi(t)])$, and the corresponding Gaussian approximation of the stationary probability distribution becomes $P_{ss}^l(\mathbf{z}, \sigma) \approx K \cdot \exp(-\phi_l(\mathbf{z})/\sigma^2)$. Here, $\mathbf{W}^+(t)$ is the pseudo-inverse matrix of the stochastic sensitivity matrix $\mathbf{W}(t)$ and these two matrices can be written as [37]:

$$\mathbf{W}(t) = \mu(t)\mathbf{P}(t) \quad \text{and} \quad \mathbf{W}^+(t) = \frac{1}{\mu(t)}\mathbf{P}(t),$$

where $\mathbf{P}(t) = \mathbf{Q}(t)\mathbf{Q}^\top(t)$ is a matrix of the orthogonal projection onto the hyperplane Θ_l . Here, $\mathbf{Q}(t)$ is a normalized vector orthogonal to $\mathbf{F}(\xi(t)) = [F_1(\xi(t)), F_2(\xi(t))]^\top$, and $\mu(t)$ is a positive T -periodic scalar function satisfies [37,38]:

$$\dot{\mu}(t) = A(t)\mu(t) + B(t), \quad \mu(0) = \mu(T), \quad (9)$$

with T -periodic coefficients $A(t) = \mathbf{Q}^\top(t)[\mathbf{J}(t) + \mathbf{J}^\top(t)]\mathbf{Q}(t)$ and $B(t) = \mathbf{Q}^\top(t)\mathbf{S}(t)\mathbf{Q}(t)$. Here, $\mathbf{J}(t)$ is the Jacobian of the deterministic system on the limit cycle $\xi(t)$, and $\mathbf{S}(t) = \text{diag}\{u^2, v^2\}$ on $\xi(t)$. The explicit solution of Eq. (9) is given by $\mu(t) = g(t)[h(t) + C]$, where

$$g(t) = \exp\left(\int_0^t A(s)ds\right), h(t) = \int_0^t \frac{b(s)}{g(s)}ds, \quad \text{and} \quad C = \frac{g(T)h(T)}{1 - g(T)},$$

and the boundaries of the confidence band can be written as:

$$[\mathbf{z} - \xi(t)]^\top \mathbf{P}(t)[\mathbf{z} - \xi(t)] = 2s^2\sigma^2\mu(t), \quad (10)$$

where $s^2 = -\ln(1 - P)$ as defined earlier.

3.3. Turing instability analysis for the noise-free spatio-temporal model

The classical Turing instability arises when a stable spatially uniform state loses stability due to a spatial perturbation, resulting in spatial patterns. Suppose, $(u, v) = (u_*, v_*)$ is the spatially homogeneous steady-state and satisfies $F_1(u_*, v_*) = 0$ and $F_2(u_*, v_*) = 0$. Furthermore, we suppose that (u_*, v_*) is locally asymptotically stable (LAS) in the absence of diffusion [39]. We substitute the spatially non-homogeneous perturbation of the homogeneous steady state of the form $u = u_* + \epsilon u_1 \exp(\lambda t) \cos kx$ and $v = v_* + \epsilon v_1 \exp(\lambda t) \cos kx$ ($\epsilon \ll 1$) into the noise-free version of the system (3), and the linearization leads to

$$\mathbf{M}_k \begin{bmatrix} u_1 \\ v_1 \end{bmatrix} \equiv \begin{bmatrix} a_{11} - d_u k^2 - \lambda & a_{12} \\ a_{21} & a_{22} - d_v k^2 - \lambda \end{bmatrix} \begin{bmatrix} u_1 \\ v_1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \quad (11)$$

where a_{11} , a_{12} , a_{21} , and a_{22} are the entries of the Jacobian matrix (5) evaluated at (u_*, v_*) . For the non-trivial solution of the system (11), $\det(\mathbf{M}_k) = 0$, which implies

$$\lambda^2 - B(k)\lambda + C(k) = 0,$$

where $B(k) = a_{11} + a_{22} - (d_u + d_v)k^2$ and $C(k) = d_u d_v k^4 - (a_{11}d_v + a_{22}d_u)k^2 + a_{11}a_{22} - a_{12}a_{21}$. According to the LAS assumption for the non-trivial steady state (u_*, v_*) , $B(k) < 0$ holds for all $k > 0$, and $C(k) > 0$ for $k = 0$. Therefore, the only condition that remains for the instability is $C(k) < 0$ for some $k > 0$, which gives in a general parametric form as:

$$(a_{11}d_v + a_{22}d_u) > 2\sqrt{d_u d_v (a_{11}a_{22} - a_{12}a_{21})}, \quad (12)$$

and the equality sign of (12) holds at $k_c = \sqrt{(a_{11}d_v + a_{22}d_u)/(2d_u d_v)}$. In addition, the positivity of the term $a_{11}d_v + a_{22}d_u$ is required for the existence of k_c . For a fixed set of temporal parameter values, if we choose the parameters d_u and d_v in such a way that the condition (12) is satisfied, then the system (3) can produce the non-homogeneous stationary pattern called the Turing pattern. Sometimes, the system produces the non-homogeneous stationary solution, but (u_*, v_*) is not locally asymptotically stable for such parameter values; in this case, the solutions are called non-homogeneous stationary patterns.

3.4. Stochastic sensitivity of pattern-attractors

Suppose, the noise-free ($\sigma = 0$) version for the system (3) has a stable stationary solution (u_*, v_*) . However, this homogeneous solution does not remain homogeneous in the presence of noise, and sufficient noise can cause instability that causes patterns in the system. Here, we find the critical noise intensity level above which the system produces patterns. For this, we perturb the stochastic system (3) as $u(x, t) = u_* + \tilde{u}(x, t) \cos(kx)$, $v(x, t) = v_* + \tilde{v}(x, t) \cos(kx)$ and expand it in a Taylor series around (u_*, v_*) [40,41]. After comparing the coefficients of $\cos(kx)$ and considering the evolving function of short wavelength $\tilde{u}_{xx} \ll k^2 \tilde{u}$ and $\tilde{v}_{xx} \ll k^2 \tilde{v}$, we obtain

$$\begin{aligned} \tilde{u}_t &= [a_{11} - d_u k^2] \tilde{u} + a_{12} \tilde{v} + \sigma \tilde{u} \xi, \\ \tilde{v}_t &= a_{21} \tilde{u} + [a_{22} - d_v k^2] \tilde{v} + \sigma \tilde{v} \eta. \end{aligned} \quad (13)$$

In discrete M spatial points in the spatial domain $[0, L]$, the stochastic differential Eq. (13) takes the form

$$\begin{aligned} \frac{d\tilde{u}_j}{dt} &= [a_{11} - d_u k^2] \tilde{u}_j + a_{12} \tilde{v}_j + \sigma \tilde{u}_j \xi_j(t), \\ \frac{d\tilde{v}_j}{dt} &= a_{21} \tilde{u}_j + [a_{22} - d_v k^2] \tilde{v}_j + \sigma \tilde{v}_j \eta_j(t), \end{aligned}$$

for $j = 1, 2, \dots, M$ and the equation of motion for the first moment become [40]:

$$\frac{d}{dt} \begin{bmatrix} \langle \tilde{u}_j \rangle \\ \langle \tilde{v}_j \rangle \end{bmatrix} = \begin{bmatrix} a_{11} + \sigma - d_u k^2 & a_{12} \\ a_{12} & a_{22} + \sigma - d_v k^2 \end{bmatrix} \begin{bmatrix} \langle \tilde{u}_j \rangle \\ \langle \tilde{v}_j \rangle \end{bmatrix} \equiv \mathbf{H}_k \begin{bmatrix} \langle \tilde{u}_j \rangle \\ \langle \tilde{v}_j \rangle \end{bmatrix}. \quad (14)$$

The stability behaviour of the homogeneous steady-state depends on the solution of the moment system (14), and it reduces to the eigenvalue problem $\det(\mathbf{H}_k - \lambda \mathbf{I}) = 0$, which gives

$$\lambda_{\pm}(k) = \frac{B(k) \pm \sqrt{B(k)^2 - 4C(k)}}{2},$$

where $B(k) = a_{11} + a_{22} - (d_u + d_v)k^2 + 2\sigma$ and $C(k) = d_u d_v k^4 - [(a_{11} + \sigma)d_v + (a_{22} + \sigma)d_u]k^2 + (a_{11} + \sigma)(a_{22} + \sigma) - a_{12}a_{21}$. Therefore, the instability of the noise system can arise if $B(k) > 0$ for some $k > 0$ or $C(k) < 0$ for some $k > 0$. According to our assumption, the noise-free system has a stable stationary solution, and it implies that $a_{11} + a_{22} - (d_u + d_v)k^2 < 0$ for all $k > 0$. However, in the presence of sufficient noise intensity σ , one can find a positive value of k such that the condition $B(k) > 0$ is true, and hence the instability can occur in such cases. On the other hand, there is a possibility for instability in the presence of a noise intensity σ when $C(k) < 0$ holds for some $k > 0$. This noise-induced instability can be determined for the system (3) similar to the Turing instability condition (12), and it gives parametric condition:

$$(a_{11} + \sigma)d_v + (a_{22} + \sigma)d_u > 2\sqrt{d_u d_v [(a_{11} + \sigma)(a_{22} + \sigma) - a_{12}a_{21}]}. \quad (15)$$

In the numerical simulation for the noise-induced patterns, we verify the instability conditions whether $B(k) > 0$ or $C(k) < 0$ or both are true for the positive k .

3.5. Noise-induced transitions between patterns

The numerical solution of the noise-free spatio-temporal model shows different types of stationary patterns in the spatial domain. The number of waves characterizes the pattern structure, and how many can fit in the domain $[0, L]$ depends on the initial conditions as it depends upon the number of unstable wavenumbers which can be accommodated within the chosen domain. Here, we consider the initial conditions as various spatial oscillatory functions in the form:

$$u(x, 0) = u_* + \epsilon \cos\left(\frac{2\pi\kappa x}{L}\right), \quad v(x, 0) = v_* + \epsilon \cos\left(\frac{2\pi\kappa x}{L}\right), \quad (16)$$

where the parameter ϵ characterizes the deviation of the initial wave from the homogeneous equilibrium, and κ is the number of pattern

Table 1

Fixed parameter values used for the numerical simulations.

Parameter	r	K	b	e	m	q
Value	1	4	0.334	0.9	0.1	0.1

peaks that can be either integer or half-integer due to the no-flux boundary conditions. In the presence of noise, these patterns do not remain stationary but oscillate in a range and preserve the pattern structure. Furthermore, depending on the initial deviation ε , a pattern structure shift is possible irrespective of the noise terms. A quantitative depiction of qualitative shifts in pattern generation is analysed by calculating the modality coefficients:

$$C_\nu(t) = \int_0^L u(x, t) \cos\left(\frac{2\pi\nu x}{L}\right) dx, \quad (17)$$

where ν can be an integer or half-integer. This coefficient C_ν contains information about the spatial wave transformation of structures in the transient process. The dominating structure has the highest absolute value of C_ν and is now the most prominent [25,29].

4. Results and discussion

This section delves into the quantitative outcomes derived from our theoretical findings. The main idea of this work is to see how environmental variability affects the dynamics of the deterministic system, particularly when the deterministic system has different types of attractors. In this work, we have considered the consumption rate a as the bifurcation parameter, and the other parameter values are fixed as given in Table 1. In addition, this section has explored the spatio-temporal behaviours for the noise-free and noise-induced models, but the temporal deterministic behaviour plays a crucial role in the species' self-organization and other patterns. So, we first explore the dynamics of the deterministic system and then discuss the spatio-temporal setup.

4.1. Deterministic dynamics

As discussed earlier, the system can have transcritical and saddle-node bifurcations, which we discuss for the fixed set of parameter values in Table 1. Fig. 1 depicts the dynamics of the deterministic system for different values of a . Fig. 1(b) is the zoom version of Fig. 1(a), which is presented to show the saddle-node bifurcations clearly. The system does not have a coexisting equilibrium point for smaller values of a , and a minimum value of a is required for the existence, which is $a_{TC} = m/e = 1/3$. In such cases, the axial equilibrium E_1 is stable, e.g., see Fig. 1(c) for $a = 0.1$. This shows that a minimum prey consumption rate is necessary for the survival of the predator population. A coexisting equilibrium point is generated through the transcritical bifurcation that occurs at $a_{TC} = 1/3$ marked by (■) in Fig. 1(a). In this case, the axial equilibrium point loses the stability, and the coexisting equilibrium point gains the stability. Two sample solutions are plotted in Fig. 1(d) for two different initial conditions.

With an increase in the parameter value of a , two saddle-node bifurcations occur, the first at $a = 0.4648$ and the second at $a = 0.4687$, and they are marked by (★) and (★) in Fig. 1(a) and (b). Two equilibrium points are generated from the first saddle-node bifurcation (★); one is a saddle point, and the other is a spiral source. In addition, the coexisting equilibrium point, generated from the transcritical bifurcation, remains a spiral sink whenever it exists. We plot the dynamics of the deterministic model after the saddle-node bifurcation for $a = 0.465$ in Fig. 1(e). In this case, one of the stable manifolds connects to the spiral source, and the unstable manifolds go to the spiral sink. In all the figures, the stable and unstable manifolds are plotted in dash-dotted and dashed curves, respectively. A further increase in the parameter value of a forms a homoclinic orbit at the saddle point, and the solutions inside the orbit converge to the spiral sink, and outside

the orbit, they go to the orbit itself [see Fig. 1(g)]. A stable limit cycle is generated from this homoclinic orbit for the higher values of a [see Fig. 1(g)–(i)]. Furthermore, two equilibrium points, the saddle and the spiral sink, disappear through the second saddle-node bifurcation marked by (★) in Fig. 1(a) and (b), and the spiral source remains the only coexisting equilibrium point for the higher values of a .

4.2. Noise-induced dynamics

The deterministic system produces two types of attractors: (i) stable equilibrium point and (ii) stable limit cycle. Here, we first discuss when a stable coexisting equilibrium point exists for the deterministic system. For the fixed parameter values given in Table 1, the system has a stable coexisting equilibrium point E_* when the value of the bifurcation parameter a lies between $1/3$ and 0.468619 [see Fig. 1(a)]. In addition, different types of dynamics occur in this parametric range due to saddle-node bifurcation. First, we focus on the scenario where the system has one stable coexisting equilibrium point. For example, we consider $a = 0.462$, and the phase portrait is shown in Fig. 1(d), and we find the confidence ellipses for different values of noise intensities. The stochastic sensitivity function approach allows for constructing confidence ellipses, a basic geometrical model for describing a configuration of random states near stable equilibrium. In addition, the stochastic sensitivity matrix provides a complete quantitative study of the stochastic sensitivity. Now, after solving Eq. (7), we obtain the stochastic sensitivity matrix as

$$\mathbf{W} = \begin{bmatrix} 6.46815 & -17.47506 \\ 25.69185 & 13.82736 \end{bmatrix},$$

and the corresponding confidence ellipse for a noise intensity σ can be obtained using (8) and it becomes

$$0.02568u^2 - u(0.01526v + 0.05919) + 0.01201v^2 - 0.03189v + 0.09689 = 2s^2\sigma^2.$$

For a low noise intensity $\sigma = 0.001$, the stochastic states of the stochastic model (2) and the confidence ellipse around the equilibrium E_* are shown in Fig. 2(a) by taking the fiducial probability $\mathcal{P} = 0.95$. It shows that the stochastic states are distributed around the stable deterministic attractor E_* , and they are situated inside the confidence ellipse, which indicates that the confidence ellipse can estimate the dispersion of the probability. Moving forward, to see how the noise intensity affects the stochastic states and the confidence ellipse, we choose two more noise intensities $\sigma = 0.01$ and 0.1 , and their results are shown in Fig. 2(b) and (c). For $\sigma = 0.01$, the stochastic states stay well inside the confidence ellipses. However, this is not true for the last case, and sometimes, the stochastic states go outside the ellipse, following a certain path, and return to the stable attractor. Therefore, the higher noise intensity predicts the possibility of having a cyclic attractor for the deterministic system. Nevertheless, Fig. 2 shows that the confidence ellipse area increases with the noise intensity. Hence, the probability dispersion increases with noise intensity, which matches the numerical simulations.

Next, we move to the case where the deterministic system has an exponentially stable attractor and two other equilibrium points that are not stable. For instance, we choose the bifurcation parameter $a = 0.467$, which lies between the two values of a in Fig. 3(e) and (f). Fig. 3(a) depicts the deterministic system's dynamics and the stable and unstable manifolds. The stochastic system can exhibit qualitative dynamics change in the presence of noise terms. The stochastic sensitivity analysis helps to predict the range of the stochastic states for a given noise intensity σ . In this case, Eq. (7) gives the stochastic sensitivity matrix as:

$$\mathbf{W} = \begin{bmatrix} 96.98917 & 3.9613 \\ 40.44592 & 20.3769 \end{bmatrix},$$

and the corresponding confidence ellipse becomes

$$0.01122u^2 + (0.02395 - 0.02445v)u + 0.0534v^2 - 0.22623v + 0.2626 = 2s^2\sigma^2,$$

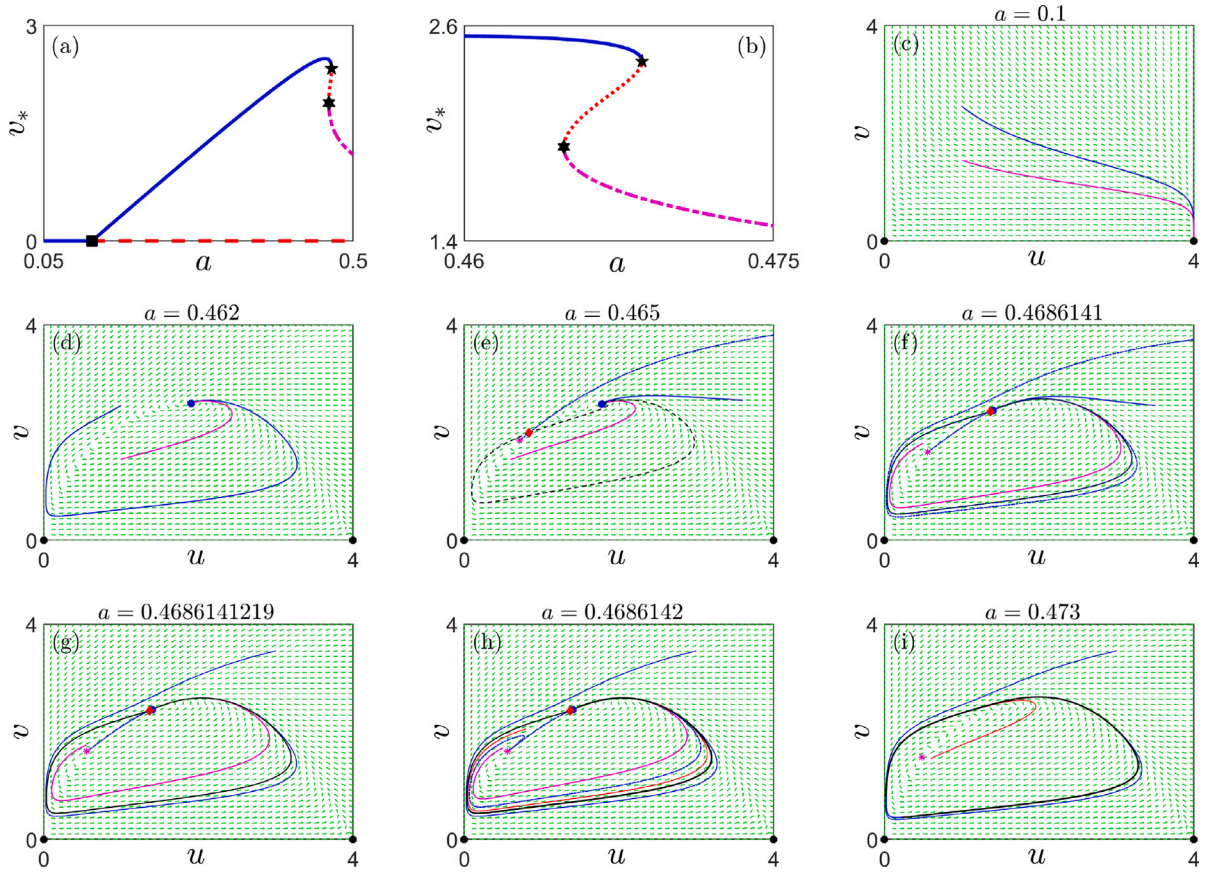


Fig. 1. Bifurcations and phase portraits of the deterministic system (1) for the bifurcation parameter a with the fixed parameter values given in Table 1. Subplots in (a) and (b) are bifurcation diagrams, and (c)–(i) are phase portraits. Subplot (b) is the zoom version of subplot (a). Transcritical bifurcation is denoted by (■), and the saddle–node bifurcations are denoted by the symbols (★) and (★). Subplots (c)–(i): vector field (green arrows), solution trajectories (solid blue, magenta, and red curves), and stable manifold (solid black curve) and unstable manifold (dotted black curve) for the saddle point for different values of a . (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

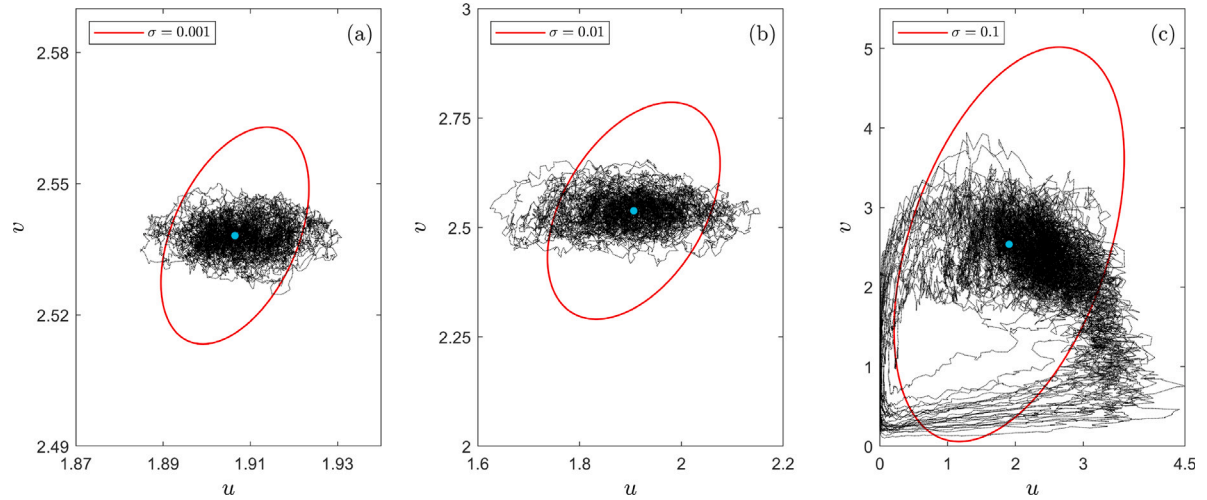


Fig. 2. The stochastic states of the stochastic model (2) for the fixed parameter values given in Table 1 with $a = 0.462$ for different noise intensities: (a) $\sigma = 0.001$, (b) $\sigma = 0.01$, and (c) $\sigma = 0.1$. The black curves are the sample trajectories, and the red curves are the confidence ellipses. The confidence ellipses (8) for the stochastic system around the stable equilibrium point for different values of σ are plotted with the fiducial probability $P = 0.95$. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

for a noise intensity σ . The confidential ellipses for the stochastic system are presented in Fig. 3(a) for three different noise intensities with the fiducial probability $P = 0.95$.

For a small noise intensity $\sigma = 0.001$, the stochastic states lie well inside the corresponding confidence ellipse [see Fig. 3(b)]. However,

they do not always stay in the confidence ellipse for $\sigma = 0.005$ and 0.01 [see Fig. 3(c) and (d)]. For $\sigma = 0.005$, the stochastic states go outside the ellipse for a while but do not go far from it. On the other hand, the stochastic states go far from the confidential ellipse for $\sigma = 0.01$ and stay close to the unstable manifold for the deterministic

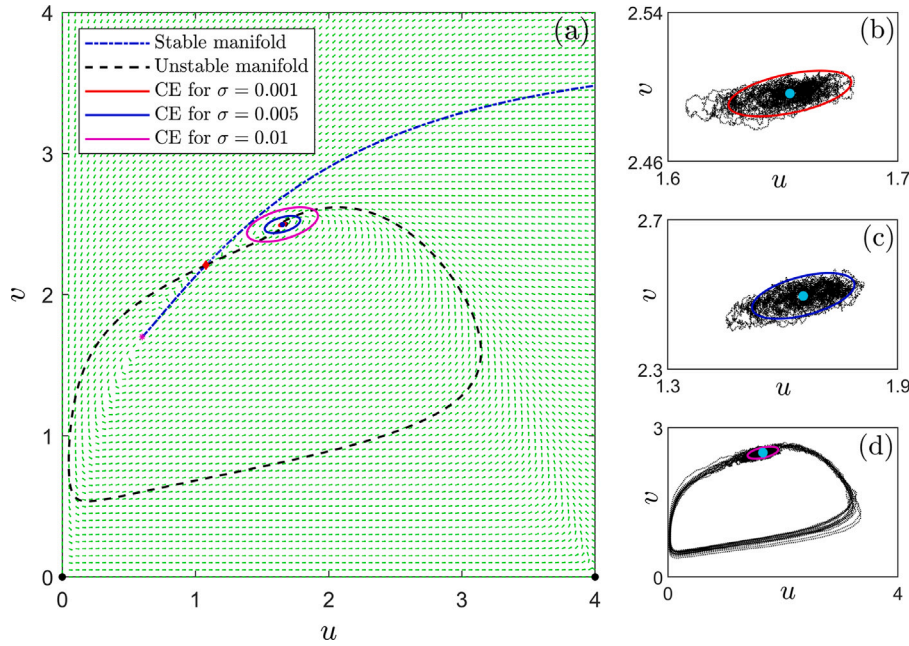


Fig. 3. (Colour online) The dynamics of the deterministic and stochastic models for the fixed parameter values are given in Table 1 with $a = 0.467$ for different noise intensities. The vector field and the equilibrium points for the deterministic system (1) are plotted in (a), and the stable and unstable manifolds for the saddle point are also presented. In addition, the confidence ellipses (CEs) for the stochastic system (2) around the stable equilibrium point are plotted in (a) for different values of σ with $P = 0.95$. The solutions of the stochastic system for $\sigma = 0.001, 0.005$, and 0.01 are plotted in (b), (c) and (d), respectively, along with their confidential ellipses.

system. The confidential ellipse crosses the stable manifold in this case, and when the stochastic states cross this stable manifold, the stochastic system makes a round outside of the unstable manifold and comes back to the confidential ellipse [see Fig. 3(a)]. In general, the deterministic attractor does not intersect stable or unstable manifolds, and the stochastic noise forces to cross the unstable manifold in this case. Therefore, the stochastic system emphasized the possibility of having a cyclic stable attractor for the deterministic system for a slide variation of the temporal parameters.

Finally, we discuss when a stable limit cycle occurs for the deterministic system, and it is the only attractor. In the presence of stochastic disturbances, the stochastic trajectories do not remain on the limit cycle but can stay around it. For a given noise intensity, we can obtain the confidential band around the limit cycle, and the random trajectories can be distributed in the band nonuniformly. In this case, the function $\mu(t)$ helps to determine the confidence band boundaries, and trajectories are confined inside them with fiducial probability P .

As mentioned earlier, two equilibrium points disappear through the second saddle-node bifurcation at $a = 0.4687$ for the fixed parameter values given in Table 1, and a stable limit cycle stays in the deterministic system for the higher values of a . This stable limit cycle (solid magenta curve) is shown in Fig. 4 for $a = 0.474$. We use Eq. (10) for finding the confidence band for the noise intensity σ . The confidence bands for the noise intensities $\sigma = 0.001, 0.003$ and 0.003 are plotted in Fig. 4 with the fiducial probability $P = 0.95$. The confidence band is narrow for $\sigma = 0.001$, and the random trajectories are concentrated near the stable limit cycle. In this case, the trajectories are completely inside the blue dashed band, and the stochastic system exhibits fairly regular oscillations. These bands expand with noise intensities, and the stochastic trajectories may not stay inside the bands for the higher intensity [see Fig. 4(c)]. Nevertheless, this type of oscillation has been predicted for the noise-induced system in earlier cases, though the stable limit cycle does not exist in such cases.

4.3. Noise-free and noise-induced patterns

We first discuss the Turing patterns in the absence of noise for the parameter values $a = 0.467$ and $d_u = 1$ with the fixed parameter

set given in Table 1. In this case, Eq. (12) helps to find the critical values $d_v^c = 17.31$ and above which the Turing instability occurs. In this case, the critical wave number becomes $k_c = 0.1738$; hence, the critical wavelength of waveforms is $w_c = 36.15$. Therefore, the spatio-temporal system (3) can produce stationary solutions with 5 to 6 population patches in the spatial domain $[0, 200]$. Here, we choose $d_v = 20$, and the computer simulations give different types of Turing patterns with different waveforms (4 to 7.5, not shown here) for different choices of the initial conditions (16) by varying κ and keeping $\epsilon = 0.001$ as fixed. Fig. 5(a) shows that the deterministic model possesses six spatial attractors for $\kappa = 3$ in the spatial domain of length $L = 200$. In this case, we calculate the modality coefficients (17) with respect to t for different values of v . Fig. 5(b) depicts such modality coefficients C_v for $v = 5.5, 6$, and 6.5 , which shows that C_6 dominates the other modalities at the end of the considered time domain $[0, 5000]$.

Next, we address the noise-induced patterns for the spatio-temporal model. We set the noise strength $\sigma = 0.01$ with the parameter values $a = 0.467$, $d_u = 1$, and the fixed parameter values given in Table 1. Here, the inequality (15) helps to find the critical diffusion coefficient for d_v and above which it is satisfied, and the critical threshold is $d_v^c = 12.89$. We fix $d_v = 14$, and its solution is shown in Fig. 5(c) for the initial condition (16) with $\kappa = 3$ and $\epsilon = 0.001$. This solution is not stationary due to the noise terms, but once the bottoms form, they are not destroyed further. It is verified that the sign of $B(k)$ remains negative for all positive k , and it shows that the instability occurs due to the sign change of $C(k)$ in a range of positive values of k . We also present the modality coefficients for the solution in Fig. 5(d), and it shows the C_5 dominates the others. It is observed that the initial conditions for both the Fig. 5(a) and (c) are the same, but the noise-induced model takes less time to generate the pattern. In addition, the noise-induced model can generate a pattern with fewer diffusion parameter values than the noise-free model, which shows that the multiplicative noise increases the chances of getting patches.

The impact of environmental variability on the noise-free spatio-temporal model is addressed when it generates the Turing patterns. It is known that the spatio-temporal model does not always generate

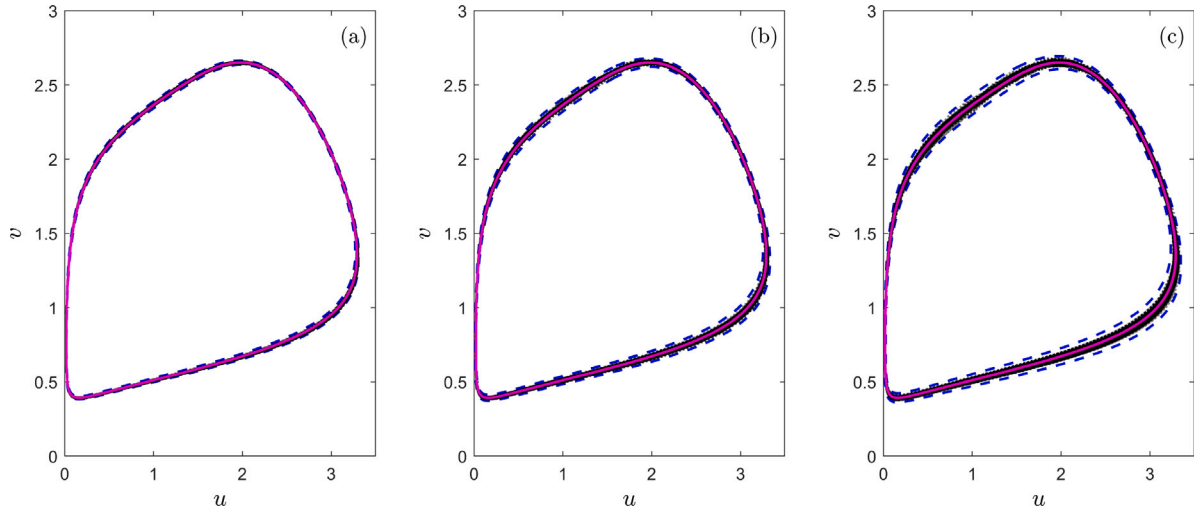


Fig. 4. The stable limit cycle (solid curve) for the deterministic system (1) and solution trajectories of the stochastic model (2) for the fixed parameter values are given in Table 1 with $a = 0.474$ for different noise intensities. The solutions of the stochastic system for $\sigma = 0.001, 0.002$, and 0.003 are plotted in (a), (b), and (c), respectively. The confidence bands (dashed curve) for the stochastic system around the stable limit cycle for different values of σ are plotted with the fiducial probability $P = 0.95$. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

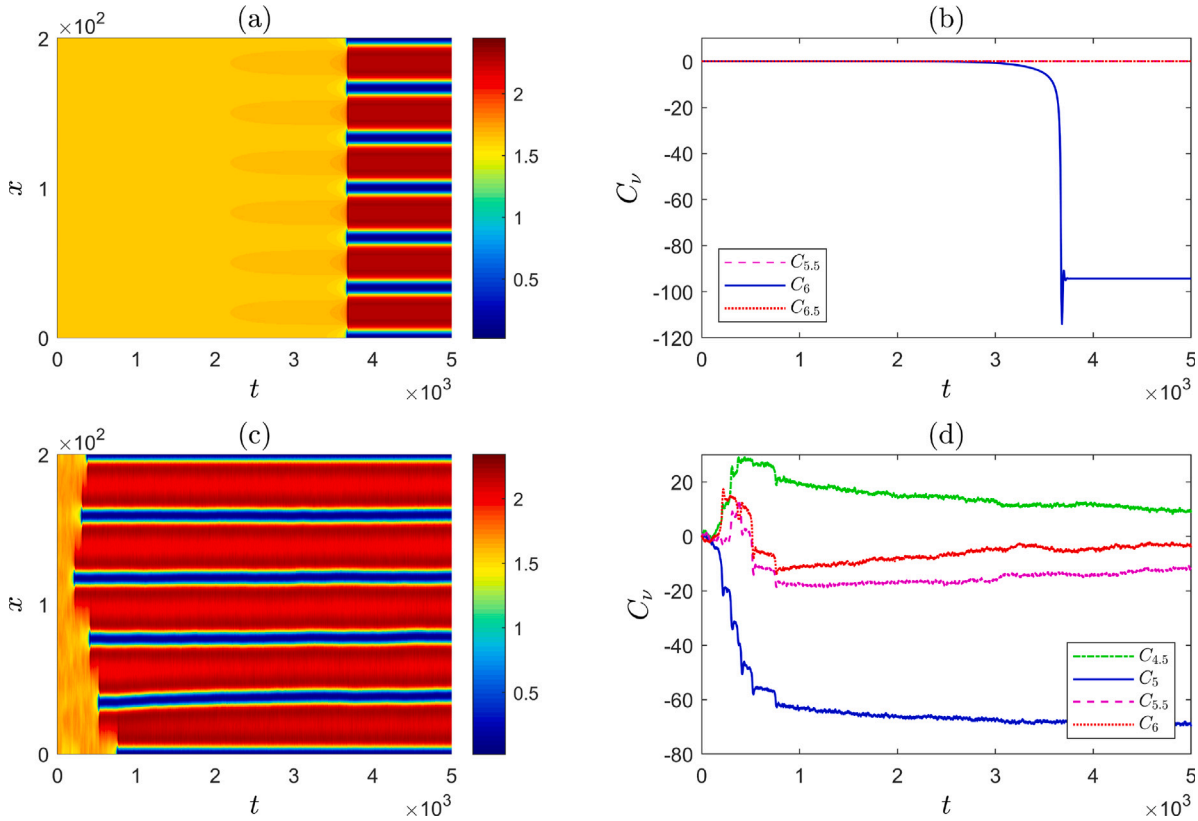


Fig. 5. (Colour online) (a) and (c) are the contour plots of the solutions corresponding to $u(x, t)$ for the spatio-temporal model (3) for the fixed parameter values given in Table 1 with $a = 0.467$, $d_u = 1$, and the initial conditions (16) with $\kappa = 3$ and $\epsilon = 0.001$. The upper panel is for $d_v = 20$, $\sigma = 0$, and the lower panel is for $d_v = 14$, $\sigma = 0.01$. The subplots (b) and (d) are the modality coefficients (C_v) for the solutions (a) and (c), respectively, and they are obtained for different values of v using (17).

such Turing patterns, even if the coexisting steady-state is locally asymptotically stable in the absence of diffusion. For a two-species model, this generally occurs when the signs of diagonal entries of the Jacobian matrix around the coexisting equilibrium point are negative. In this case, we cannot find the diffusion coefficients d_u and d_v such that the condition (12) holds. However, the multiplicative noise can help to generate some stationary patches called noise-induced patterns, and we present one such case.

We fix $a = 0.462$, and the coexisting equilibrium point E_* is stable. In this case, the diagonal entries for its Jacobian matrix are negative, and the Turing instability analysis cannot apply to the noise-free spatio-temporal model. However, we cannot find the diffusion parameter values for the instability of the noise-induced model with sufficiently small noise σ . This happens because there do not exist any combinations of d_u and d_v such that the inequality (15) holds. Therefore, a minimum noise intensity level is required for the instability such that

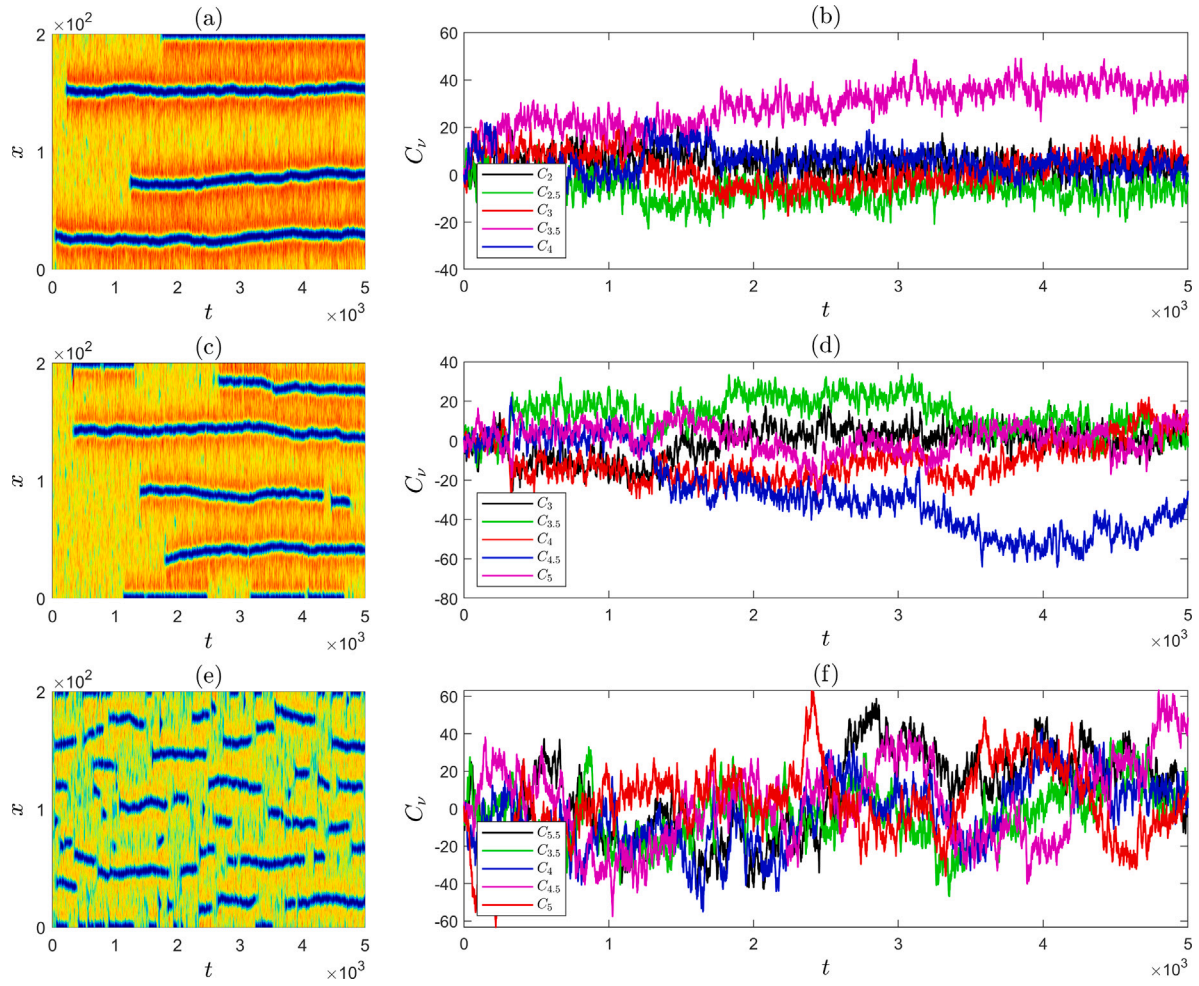


Fig. 6. (Colour online) (a), (c), and (e) are the contour plots of the solutions corresponding to $u(x, t)$ for the spatio-temporal model (3) for the fixed parameter values given in Table 1 with $a = 0.467$, $d_u = 1$, and the initial conditions (16) with $\kappa = 3$ and $\epsilon = 0.001$. The upper panel is for $d_v = 25$, $\sigma = 0.1$, the middle panel is for $d_v = 15$, $\sigma = 0.1$, and the lower panel is for $d_v = 15$, $\sigma = 0.15$. The subplots (b), (d), and (f) are the modality coefficients (C_v) for the solutions (a), (c), and (e) respectively.

either of the signs $a_{11} + \sigma$ or $a_{22} + \sigma$ in (15) becomes positive, and this minimum value varies with the background temporal parameter values.

We first choose $\sigma = 0.1$ so the term $a_{11} + \sigma$ becomes positive. For $d_u = 1$, the critical diffusion threshold turns out to be $d_v^c = 18.62$, and above it, the inequality (15) holds. Fig. 6(a) depicts the noise-induced patterns for $d_v = 25$, and the modality coefficient $C_{3.5}$ dominates the others [see Fig. 6(b)]. In addition, the sign of $B(k)$ remains negative for all positive k . It is also observed that the noise-induced spatio-temporal model produces self-organization patches for the diffusion parameter values smaller than the critical threshold, but those patches may not sustain for a longer time. Fig. 6(c) shows such type of long-transient dynamics for $d_v = 15 < d_v^c$, and this solution exists because of the oscillations predicted in the noise-induced temporal model [see Fig. 2(c)]. Furthermore, we have presented the modality coefficients for the solution in Fig. 6(d), which shows the transitions of patterns in the considered time domain. Nevertheless, the noise-free spatio-temporal model predicts only the sharp values for the diffusion parameters for which the Turing pattern occurs. However, this is not always true in ecosystems because environmental variability affects those critical values and results in long transients in the system.

Sometimes, noise can cause irregular oscillations in the patterns, and there is no clear indication of the number of patches in a spatial domain. Fig. 6(e) depicts one of such solution behaviours for the noise-induced spatio-temporal model for $\sigma = 0.15$, $d_u = 1$, and $d_v = 15$. In this case, the system shows nonequilibrium dynamics where it does not settle into a stable stationary state or a periodic cycle. The evolution

of modality coefficients for this dynamic pattern shows the number of patches shifting from one to another [see Fig. 6(f)], and no dominance behaviour is observed. The inequality (15) is satisfied for the instability in this case, but the term $B(k)$ remains positive for a range of positive values of k , which causes spatio-temporal oscillation those generally observed for the noise-free model in the Hopf unstable domain. Therefore, the noise-induced spatio-temporal model shows self-organization patches in a wider range of diffusion parameter values and produces different patterns depending on the level of environmental variability.

5. Conclusions

This paper is devoted to a mathematical framework of predator-prey interactions in ecological systems, considering the influence of linear random disturbances through multiplicative noise. The trajectories of stochastic attractors may not always stay near the basin of attraction of the deterministic system. However, it depends on the noise intensity and position of other nearby attractors. The stochastic sensitivity approach shows that the stochastic attractors are well inside the confidential ellipses for lower noise intensities; however, the trajectories go outside the ellipse for the higher intensities and return after making rounds. These behaviours are observed when the deterministic system has a stable equilibrium attractor in the presence and absence of other unstable equilibrium points. Therefore, the noise-induced system predicts that the deterministic system may have a stable orbit that can be found by tuning the background parameter values.

According to the predictions, the stable limit cycle is observed by changing the bifurcation value. The stochastic sensitivity analysis helps to construct the confidence bands in the neighbourhood of the limit cycle for a certain probability for different noise intensities. The results show that the width of the band increases with the noise intensity, but the trajectories remain well inside the confidence bands. This indicates that the multiplicative noise cannot change the dynamics of the deterministic system when it exists the oscillatory solutions. Therefore, the oscillatory dynamics in the deterministic system can absorb the noise due to linear order random disturbances that appear from the population's growth and death rates. Nevertheless, these analyses help in predicting the survival of the populations under environmental fluctuations.

This work also studies the noise-induced spatio-temporal model. It is known that the noise-free spatio-temporal model can produce Turing patterns, but some parametric conditions are required. Here, we have shown that the noise-induced model can help to form patches in less time and a much wider range of parameter values. Sometimes, the noise-free spatio-temporal model cannot generate spatial patterns. However, environmental variability influences the production of patches in the spatial domain and requires minimum strength. In addition, it is observed that the higher noise intensity can force the spatial system into a nonequilibrium state where new patches form and are destroyed after some time. Furthermore, the self-organization behaviour is observed in a wider range of parameter values for the noise-induced model compared to the noise-free model. This study covers the noise-induced dynamics in a one-dimensional spatial domain and could be used to analyse the spatio-temporal behaviours of populations whose dynamics are strongly affected by the noise. It would be fascinating to see the stochastic fluctuations in a two-dimensional spatial domain when the model generates stationary patches and when it does not. Furthermore, it would be interesting to investigate higher-order random disturbances in growth rates rather than linear order or disturbances of the parameters involved in predation and other factors. The feedback control strategy can be applied to keep the corresponding confidence ellipse smaller [26], and the noise-induced system can resist a larger noise intensity before going for the large oscillations.

CRediT authorship contribution statement

Swadesh Pal: Writing – original draft, Visualization, Validation, Resources, Methodology, Investigation, Formal analysis, Data curation, Software. **Malay Banerjee:** Writing – review & editing, Visualization, Supervision, Methodology, Conceptualization. **Roderick Melnik:** Writing – review & editing, Supervision, Project administration, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

No data was used for the research described in the article.

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